

Assignment-1

Q1.Find area of rectangle.

```
length = float(input("Enter length: "))
breadth = float(input("Enter breadth: "))

area = length * breadth

print("Area of Rectangle =", area)

Enter length: 5
Enter breadth: 4
Area of Rectangle = 20.0
```

Q2.Find simple interest.

```
principal = float(input("Enter Principal Amount: "))
rate = float(input("Enter Rate of Interest: "))
time = float(input("Enter Time (in years): "))

simple_interest = (principal * rate * time) / 100

print("Simple Interest =", simple_interest)

Enter Principal Amount: 100000
Enter Rate of Interest: 12
Enter Time (in years): 10
Simple Interest = 120000.0
```

Q3.Convert temperature from Celcius to Fahrenheit.

```
celsius = float(input("Enter temperature in Celsius: "))

fahrenheit = (celsius * 9/5) + 32

print("Temperature in Fahrenheit =", fahrenheit)

Enter temperature in celsius: 100
Temperature in Fahrenheit = 212.0
```

Q4.Calculate average of 3 numbers.

```
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
num3 = float(input("Enter third number: "))

average = (num1 + num2 + num3) / 3

print("Average =", average)

Enter first number: 5
Enter second number: 8
Enter third number: 11
Average = 8.0
```

Q5.Find square and cube of a number.

```
num = float(input("Enter a number: "))

square = num ** 2
cube = num ** 3

print("Square =", square)
print("Cube =", cube)

Enter a number: 15
Square = 225.0
Cube = 3375.0
```

Q6.Swap two numbers without third variable.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Before Swapping:")
print("a =", a)
print("b =", b)

a = a + b
b = a - b
a = a - b

print("After Swapping:")
print("a =", a)
print("b =", b)

Enter first number: 5
Enter second number: 10
Before Swapping:
a = 5
b = 10
After Swapping:
a = 10
b = 5
```

Q7. Create a Student Report Program that take student details using input(), Store marks in variables, Calculate total and percentage , Use comments, Use proper indentation.

```
# Student Report Program

# Taking student details
name = input("Enter Student Name: ")
roll_no = input("Enter Roll Number: ")

# Taking marks of 5 subjects
mark1 = float(input("Enter marks in Subject 1: "))
mark2 = float(input("Enter marks in Subject 2: "))
mark3 = float(input("Enter marks in Subject 3: "))
mark4 = float(input("Enter marks in Subject 4: "))
mark5 = float(input("Enter marks in Subject 5: "))

# Calculating total marks
total = mark1 + mark2 + mark3 + mark4 + mark5

# Calculating percentage
percentage = total / 5
```

```
# Displaying report
print("\n----- STUDENT REPORT -----")
print("Student Name :", name)
print("Roll Number :", roll_no)
print("Total Marks :", total)
print("Percentage :", percentage, "%")

# Determining Grade
if percentage >= 90:
    grade = "A+"
elif percentage >= 80:
    grade = "A"
elif percentage >= 70:
    grade = "B"
elif percentage >= 60:
    grade = "C"
elif percentage >= 50:
    grade = "D"
else:
    grade = "Fail"

print("Grade      :", grade)
```

```
Enter Student Name: Astha
Enter Roll Number: 5
Enter marks in Subject 1: 95
Enter marks in Subject 2: 99
Enter marks in Subject 3: 97
Enter marks in Subject 4: 92
Enter marks in Subject 5: 98

----- STUDENT REPORT -----
Student Name : Astha
Roll Number  : 5
Total Marks  : 481.0
Percentage   : 96.2 %
Grade       : A+
```